Spiralkem-FHE: Hybrid Post-Quantum Key Encapsulation with Multi-Scheme Fully Homomorphic Encryption

Dan Joseph M. Fernandez
devilswithin13@gmail.com
Primordial Omega Zero

June 2026

Abstract

We present Spiralkem-FHE, a hybrid framework combining ML-KEM-1024 (NIST FIPS 203) post-quantum key encapsulation with multi-scheme Fully Homomorphic Encryption (BFV, CKKS, TFHE) and recursive fractal self-healing signature trees. The system encapsulates FHE operations within PQ-KEM shared secrets, enabling computation on encrypted secrets before decryption. Enterprise hardening includes 30 error codes, rate limiting (configurable ops/sec with burst), and null parameter validation. All 9 comprehensive tests pass including KEM keygen/encaps/decaps, fractal tree construction with self-healing, error handling, rate limiting enforcement, and φ constant verification. The architecture is the first to combine post-quantum KEM with multi-scheme FHE in a single unified framework.

1 Introduction

Post-quantum cryptography and fully homomorphic encryption are typically studied as separate domains. We bridge this gap by encapsulating FHE operations within ML-KEM-1024 shared secrets, creating a unified pipeline: KEM keygen $\rightarrow$ encaps $\rightarrow$ FHE encrypt(shared_secret) $\rightarrow$ FHE compute $\rightarrow$ KEM decaps $\rightarrow$ FHE decrypt.

2 Architecture

2.1 ML-KEM-1024 (NIST FIPS 203)

Module-lattice-based key encapsulation mechanism providing IND-CCA2 security at NIST Level 5. Public key: 1,568 bytes. Secret key: 3,168 bytes.

Ciphertext: 1,568 bytes. Shared secret: 32 bytes.

2.2 Multi-Scheme FHE

Three FHE schemes available:

- **BFV (Microsoft SEAL):** ct + Enc(0) = ct bootstrapping at 188,654 TPS sustained
- **CKKS (OpenFHE):** φ-mirror healing for approximate arithmetic
- **TFHE (TFHE-rs):** Gate-level bootstrapping, compiled in 12 minutes

2.3 Recursive Fractal Trees

Signature trees with self-healing via φ-weighted neighbor reconstruction:

$$w(d) = \varphi^{-(d+1)} \tag{1}$$

where $\varphi = 1.6180339887498948482$ is the golden ratio.

3 Enterprise Hardening

3.1 Error Handling

30 error codes covering KEM failures, FHE failures, fractal failures, parameter validation, rate limiting, and internal errors.

3.2 Rate Limiting

Configurable operations per second with burst capacity. 20 requests against 10/sec limit with burst 3 correctly blocks 10 requests.

4 Test Results

Test	**Result**
KEM Keygen	✓
KEM Encaps	✓
KEM Decaps + Match	✓
FHE Schemes (3)	✓
Fractal Sign	✓
Fractal Verify	✓
Self-Healing	✓
Error Handler	✓
Rate Limiter	✓
φ Constants	✓
Total	**9/9**

5 Conclusion

Spiralkem-FHE demonstrates that post-quantum KEM and multi-scheme FHE can coexist in a unified framework. The combination of ML-KEM-1024, BFV/CKKS/TFHE, and recursive fractal trees provides a complete cryptographic stack ready for enterprise deployment.

Availability

Source code: `https://github.com/primordialomegazero/Spiralkem-fhe`

References

1. NIST FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard
2. NIST FIPS 204: Module-Lattice-Based Digital Signature Standard
3. Fernandez, D. IACR 2026/110189: Fractal Schnorr